

PRATIK MANDANI

PRODUCT DEVELOPMENT & FULL-STACK ENGINEER

Web Architecture (PHP / Laravel) • Embedded Systems & Firmware (ESP32 / C/C++)

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PROFESSIONAL SUMMARY

Engineer with **7 years of hands-on experience** spanning vending-machine product development, electronics, hardware integration, machine testing, R&D, and web application development. Currently working with PHP/Laravel delivering **100+ commercial websites and web platforms**, including Core Web Vitals optimization achieving **99/100 Lighthouse performance** and 0ms blocking time. Actively developing embedded IoT firmware projects with ESP32 and C/C++.

TECHNICAL SKILLS

Web & Backend	PHP, Laravel, Node.js, Express.js, MySQL, RESTful APIs, MVC Architecture, Database Design
Frontend & UI	JavaScript, Tailwind CSS, Bootstrap, HTML5, CSS3, Responsive Design, Figma to Code
Embedded & Firmware	ESP32 Dual-Core, C/C++, Arduino IDE, ESP-IDF, I2C, I2S DMA, ADC, GPIO, LittleFS, State Machines
AI & Speech Pipelines	Google Gemini (Free API Tokens), Groq LLaMA 3.3, Groq Whisper-v3 STT, Piper Neural TTS (ONNX), FFmpeg
Hardware & Electronics	Proteus 8 CAD, PCB Assembly, Wire Harnessing, LM2596 Regulators, PC817 Optocouplers, Testing
Tools & Optimization	Core Web Vitals, Google Lighthouse (99 Desktop / 93 Mobile), Git, GitHub, Linux, Apache

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer — Altius Infoway	May 2022 – Present (4 Years)
- Architected and deployed over 100 websites and web applications across corporate, business, e-commerce, and management applications with Laravel and MySQL. - Directed hands-on frontend styling, custom Laravel development, database integration, maintenance, and production deployment. - Built role-based administrative panels, CRM lead tracking modules, and secure checkout workflows. - Audited and tuned Core Web Vitals to achieve a verified 99/100 Desktop and 93/100 Mobile Lighthouse Performance score with 0ms Total Blocking Time.	
Product Development & Electronics Engineer — Symbian Vending Solutions	Nov 2018 – Mar 2022 (3 Years)
- Executed machine assembly, internal wiring, harness fabrication, and hardware integration for commercial beverage and cup-dispensing vending machines. - Conducted systematic machine testing, operational validation, issue identification, and R&D support across diverse vending prototypes. - Facilitated hardware-firmware coordination with development teams to resolve electrical noise and operational timing constraints. - Supported engineering of reusable vending control cards, 24V-12V coin interface modules, and automatic cup dispensing mechanisms.	

KEY PROJECTS & HARDWARE IMPLEMENTATIONS

The Busa — Fashion Brand E-Commerce Platform

Laravel • MySQL • JavaScript

- Engineered full e-commerce platform with product catalog, cart system, and order management.
- Optimized asset delivery and caching to achieve verified **99 Desktop / 93 Mobile Lighthouse Score** with 100 Best Practices and 0ms TBT.
- Live: thebusa.com

KOTL Assistant Robot — Desktop AI Voice Robot

ESP32 • Node.js • Gemini AI • Whisper • Piper

- Built real-time conversational robot connecting ESP32 edge hardware with a Node.js REST backend over Wi-Fi HTTP.
- Implemented 8kHz PCM voice capture (MAX9814 AGC Mic), multilingual Groq Whisper-v3 STT, free Google Gemini AI dialogue reasoning, and local Piper neural TTS (16kHz WAV) streamed to a MAX98357A I2S DAC amp with synced OLED RoboEyes facial expressions.

Diamorivo — Luxury Fine Jewelry E-Commerce

Laravel • MySQL • Tailwind CSS

- Implemented high-converting jewelry catalog and checkout workflow from Figma designs, ensuring complete mobile responsiveness.
- Live: diamorivo.com

Commercial Vending Machine Electronics & Subsystems

Proteus 8 CAD • Power Regulation • Relays

- **Reusable Control Card:** Standardized multi-recipe dispensing across diverse tea/coffee machine models.
- **24V–12V Coin Interface Module:** LM2596 step-down regulation with PC817 optocoupler isolation to protect control lines.
- **Automatic Cup Dispensing:** Validated motor actuators, touch interfaces, and QR payment integration under operational load.

EDUCATION

Diploma in Electronics & Communication Engineering (EC)

GTU Certified

Government Polytechnic, Rajkot — Gujarat Technological University (GTU)